MAT 181 Chapter 8 Continues Practice Problems-2-Ans
Name_____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.
Identify the null hypothesis, alternative hypothesis, test statistic, P-value, conclusion about the null hypothesis, and final conclusion that addresses the original claim.

1) A manufacturer considers his production process to be out of control when defects exceed 3%. In a random sample of 85 items, the defect rate is 5.9% but the manager claims that this is only a sample fluctuation and production is not really out of control. At the 0.01 level of significance, test the manager's claim. 1) _____

2) A poll of 1068 adult Americans reveals that 48% of the voters surveyed prefer the Democratic candidate for the presidency. At the 0.05 level of significance, test the claim that at least half of all voters prefer the Democrat. 2) _____

3) According to a recent poll 53% of Americans would vote for the incumbent president. If a random sample of 100 people results in 45% who would vote for the incumbent, test the claim that the actual percentage is 53%. Use a 0.10 significance level. 3) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Find the P-value for the indicated hypothesis test.

4) A medical school claims that more than 28% of its students plan to go into general practice. It is found that among a random sample of 130 of the school's students, 32% of them plan to go into general practice. Find the P-value for a test of the school's claim. 4) _____

A) 0.1539 B) 0.3461 C) 0.1635 D) 0.3078

5) In a sample of 88 children selected randomly from one town, it is found that 8 of them suffer from asthma. Find the P-value for a test of the claim that the proportion of all children in the town who suffer from asthma is equal to 11%. 5) _____

A) -0.2843 B) 0.2843 C) 0.2157 D) 0.5686

6) A manufacturer claims that fewer than 6% of its fax machines are defective. In a random sample of 97 such fax machines, 5% are defective. Find the P-value for a test of the manufacturer's claim. 6) _____

A) 0.3264 B) 0.1591 C) 0.3409 D) 0.1736

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Assume that a simple random sample has been selected from a normally distributed population. Find the test statistic, P-value, critical value(s), and state the final conclusion.

7) Test the claim that for the population of female college students, the mean weight is given by $\mu = 132$ lb. Sample data are summarized as $n = 20$, $\bar{x} = 137$ lb, and $s = 14.2$ lb. Use a significance level of $\alpha = 0.1$.

7) _____

8) Test the claim that for the adult population of one town, the mean annual salary is given by $\mu = \$30{,}000$. Sample data are summarized as $n = 17$, $\bar{x} = \$22{,}298$, and $s = \$14{,}200$. Use a significance level of $\alpha = 0.05$.

8) _____

Assume that a simple random sample has been selected from a normally distributed population.
Find the test statistic, P-value, critical value(s), and state the final conclusion.

9) Test the claim that for the population of female college students, the mean weight is given by $\mu = 132$ lb. Sample data are summarized as $n = 20$, $\bar{x} = 137$ lb, and $s = 14.2$ lb.
Use a significance level of $\alpha = 0.1$.

9) _____